PS 3

ARE 212, Spring 2026

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```r
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(haven)

path <- "~/projects/ARE212_2026/"
path_ps3 <- file.path(path, "ps3")
path_data <- file.path(path_ps3, "data")

ggplot2::theme_set(
  theme_bw() +
    theme(
      text = element_text(family = "sans"),
      strip.background = element_blank()
    )
)
```

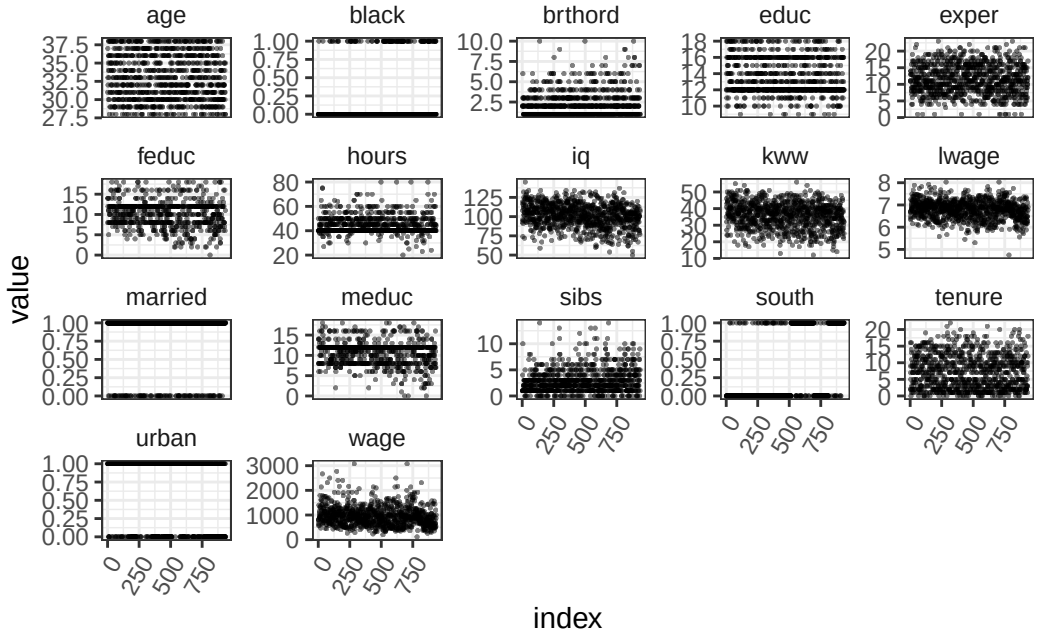
Q1

Read data in and plot series to check data quality.

```
data_nls80 <- read_dta(file.path(path_data, "NLS80.dta"))
```

```
data_nls80 %>%
  mutate(index = row_number()) %>%
  pivot_longer(
    cols = !index,
    names_to = "var"
  ) %>%
  ggplot(aes(index, value)) +
  geom_point(
    shape = 21,
    alpha = 0.5,
    fill = "black",
    stroke = 0,
    size = 0.75
  ) +
  facet_wrap(~var, scales = "free_y") +
  theme(
    axis.text.x = element_text(angle = 60, hjust = 1)
  )
```

```
Warning: Removed 355 rows containing missing values or values outside the scale range
(`geom_point()`).
```

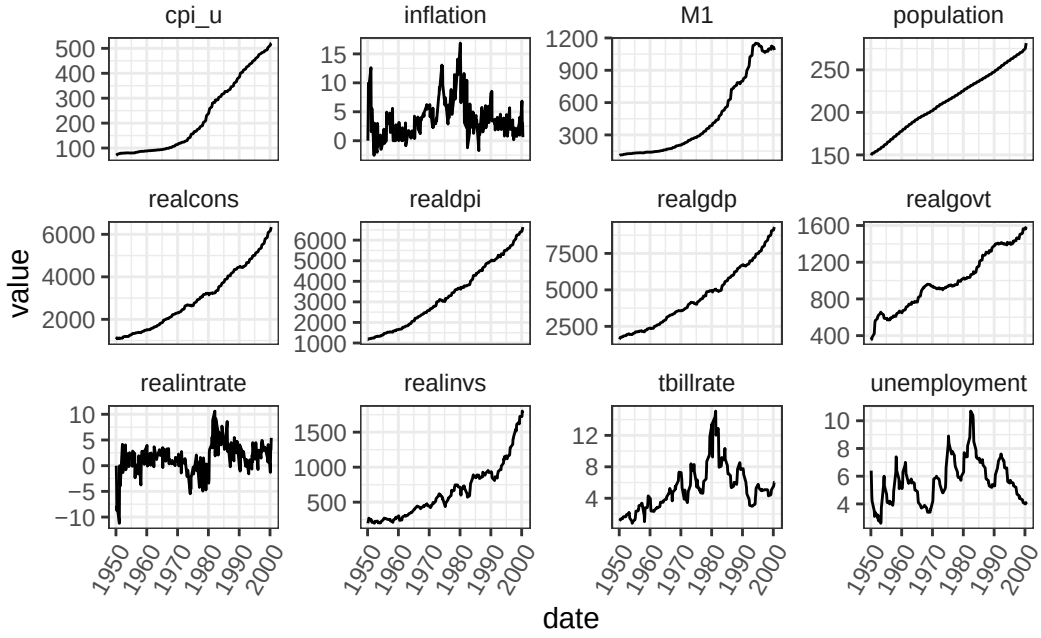


Q2

Q3

```
data_macro <- read_dta(file.path(path_data, "macrodata.dta"))
```

```
data_macro %>%
  mutate(date = year + (qtr / 10)) %>%
  select(!c(year, qtr)) %>%
  pivot_longer(
    cols = !date,
    names_to = "variable"
  ) %>%
  ggplot(aes(date, value, group = variable)) +
  geom_line() +
  facet_wrap(~variable, scales = "free_y") +
  theme(
    axis.text.x = element_text(angle = 60, hjust = 1)
  )
```


cpi_u
inflation
M1
population
realcons
realdpi
realgdp
realgovt
realintrate
realinvs
tbillrate
unemployment
value
date

Q4

Q5